Technical Report

Submission-to-Decision Traceability Platform

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1 Scope Alignment and Feature Validation

The approved workflow is implemented end to end. Citizens send views on tracked bills and receive hash-chained receipts. Committee reports are read and split into entries. An AI proposes which entry answers each view, and a moderator confirms or rejects every proposal before any outcome is shown. This release extends the trace beyond the committee report, which was the earlier stopping point. Each submission recorded in a report can now be followed to what the House decided in the Committee of the Whole House, and to whether it appears in the final Act. A citizen's own receipt joins that trace once a moderator confirms its match. Coverage is the National Assembly and the two pilot counties, Nairobi City and Trans Nzoia. **Status** below is local validation; this release is not yet deployed (Section 3, item 4).

Planned feature	Purpose	Status	Evidence / caveat
National Assembly tracking	Bills, stage and dated history from Parliament's Bills Tracker; the bill text itself.	Implemented, tested	`sync_legislation.py`: 714 bills in total. The sync now attaches each bill's PDF from Parliament's Bills pages: 260 of 409 National Assembly bills (proposals excluded) have their text.
County assemblies	Nairobi and Trans Nzoia bills, their Acts on Kenya Law, and county committees.	Implemented, tested	`sync_county_bills.py`, `committee_assignment.py`: 104 + 4 records, 28 committees. Assemblies publish no bill stages, so none are invented.
Participation and submission	Accept views only while a committee takes them; one view per bill, editable; memoranda upload.	Implemented, tested	Only the "Public participation" stage accepts views. The sync sets it while a call for memoranda is open and moves the bill on when the call closes; a clerk recording a call does the same. On 13 November no call was open (the newest closed on 31 August), so 0 bills accept views.
Receipt integrity	Append-only SHA-256 chain; edits as new versions; full and single-receipt verification.	Implemented, tested	`receipt_service.py`, database triggers. Hosted chain (7 November): 10 of 10 receipts verify after the demo-data cleanup.
Calls for memoranda	Parliament's calls, with window, mode and bill-text access.	Implemented, tested	35 notices. Shown as a status on each bill with a "Has a call for views" filter; the separate notices page was retired.
Matching and review	Propose which report entry answers each platform view; a moderator decides.	Implemented; 82% accuracy on 60 labelled cases	`llm_matching_service.py`, `ModeratorPages.tsx`. Precision 91–97%, recall 78–80% (text-similarity baseline: 15% and 18%).
What each report records	Who made each submission in a committee report, what they asked, and the committee's decision.	Implemented, tested	`report_analysis_service.py`: 648 submissions in 51 reports. 79 names recovered from merged table cells; 106 (16%) stay "Not named in the report" because the report never attributes them, with the stakeholders it lists shown instead.
What the House decided	Each adopted recommendation's fate in the Committee of the Whole House.	Implemented, tested; 10 bills	`house_trace_service.py`, Hansard via Mzalendo. 176 recommendations: 91 agreed, 1 negatived, 46 change not made (clause passed unchanged or no amendment), 37 could not confirm. No debate found for 8 passed bills, mostly Senate or older sittings.
Whether it became law	Check each adopted recommendation against the final Act; read the Act with those passages highlighted.	Implemented, tested; 12 Acts	`act_trace_service.py`. 82 bills linked to their Act on Kenya Law; 12 Act texts loaded. 203 recommendations: 113 in the Act, 24 not, 66 could not confirm. Act reader at `/bills/:id/act`.
Receipt to law	A citizen's receipt follows its matched report entry through the House and into the Act.	Implemented, tested	Receipt timeline "From this view to the law"; matched report entries marked "Also sent on this platform". Needs a confirmed match, so no hosted receipt shows it yet.
Roles, stages, offline	Role home pages; Clerk's Office limited to one committee; abuse limits; plain-language stages; offline drafts.	Implemented, tested	`ensure_committee_scope`, `rate_limit.py`, `submission_stage.py`, `utils/offline.ts`; `test_security.py`.

Outside this release: forwarding submissions to committee clerks (so the platform cannot yet prove delivery); a moderator sign-off for the AI's report, House and Act readings (they are checked automatically, but no person confirms them); automatic Act texts; Senate and older House debates that Mzalendo does not hold; an interface in Kiswahili (notifications are bilingual); and assemblies beyond the two pilot counties.

2 Technical Functionality and Architecture

A React/TypeScript (Vite) client on Netlify calls a FastAPI REST API on Render backed by PostgreSQL on Neon. Requests pass an in-memory rate limiter, JWT authentication and role checks; Clerk's Office actions also pass a committee-scope check. Routes delegate to domain modules: `participation` (when a bill accepts views, and the stage a call for memoranda implies), `submission_stage`, `receipt_service`, `ingestion_service` (`pdftotext` with Tesseract OCR fallback), `linking_service` and `llm_matching_service` (Gemini proposals), and three new tracing services: `report_analysis_service`, `house_trace_service` and `act_trace_service`. A shared `evidence` module supports all three. Sync scripts read official sources and accept only same-domain links. Each AI step runs from a script or a staff action, never from a page view. Results are stored in `report_analyses`, `house_traces` and `act_traces` (four new migrations).

Sources. Bills, stages, reports, calls for memoranda and bill PDFs come from parliament.go.ke. House debates come from Mzalendo's copy of the Hansard, read politely and credited on every page that uses it; robots.txt allows this, and a note to Mzalendo about reuse is pending. Kenya Law serves Act pages behind an automated-traffic challenge. The platform therefore reads only its public listing, to link each bill to its Act, and never downloads Act pages. Act texts come from PDFs that staff download in a browser and upload, or from Parliament's own Acts page. The Laws.Africa API would automate this but is paid.

How AI evidence is checked. Gemini never supplies the evidence itself. It receives the Act, bill or Hansard with numbered lines and cites a line range; the platform takes the wording from those lines. A claim stands only if checks the platform runs itself agree:

- **Act:** the cited Act wording must not already be in the bill as published (clause numbers ignored), unless the stakeholder supported that wording. A deletion must name bill wording that is missing from the Act.
- **House:** the cited lines must record an amendment being moved and the decision, on the clause the recommendation concerns. "(Clauses 29–32 agreed to)" counts as **passed unchanged**, never as an agreed change.
- **Second check:** a separate Gemini call sees only the word-level difference, or the amendment moved, and must agree that it carries out the recommendation.

Answers that fail a check are retried once with the reason; anything still unproven shows as "could not confirm". Stakeholder names are accepted only if they appear in the report text shown to the model. Receipts keep their earlier protections: triggers reject UPDATE and DELETE, edits are new versions, and hashes cover uploaded documents.

3 Testing Approaches and Results

1. Automated checks. `npm run typecheck` and `npm run build` pass. The main bundle is 1,370 kB (397 kB gzip), above Vite's 500 kB advisory. `npm audit` reports 5 advisories (3 moderate, 2 high) in react-router, vite, uuid and source-map-js; the React Router open-redirect case is mitigated in the sign-in redirect helper. The backend suite gives **228 passed** with **88% line coverage**. That is down from 97%: the three new command-line scripts (`trace_acts`, `trace_house`, `analyse_reports`) are untested, and so is the Hansard-fetching half of `house_trace_service` (56%); routes stay at 98%. Tests run against PostgreSQL with the production append-only triggers. Gemini, HTTP and PDF tools are faked, so tests never go online. Beyond the earlier areas (tamper detection, participation windows, committee scope, rate limits, AI-score capping, the tracker and county parsers), new tests cover:

- the participation stage following calls for memoranda, and a clerk's call opening a First Reading bill;
- the evidence checks: invented quotes, renumbered unchanged wording, supported wording, deletions, OCR misreadings ("issucd" for "issued") but never a changed number ("ten" for "five"), and the second check rejecting an unrelated change;
- House rules: wrong clause, no amendment moved, and "passed unchanged";
- name recovery rejecting a name that is not in the report;
- a receipt following its report entry through the House into the Act.

2. Browser checks (local, Playwright, Chromium). The bill page, the Act reader, the receipt timeline and the Bills filters rendered with no console errors. A defect was found and fixed: when two submissions pointed to the same Act wording, the reader listed the second under "could not confirm" although its badge said "In the final Act".

3. Matching and usability (unchanged since 10 November). Matching scored 82% accuracy over three runs on 60 labelled submissions against three real reports (precision 91–97%, recall 78–80%; baseline 15% and 18%). Every proposal scored 0.9 or above was correct; Kiswahili was weakest (1 of 3). The usability study of 35 adults gave 80% mean task completion and a mean SUS of 72.4 (68 benchmark). Finding the committee outcome was hardest (60%). This release addresses that with the receipt timeline and the per-submission trace on bill pages.

3a. Traceability checks (13 November 2026). The House and Act checks ran on every passed bill with an analysed report and on the 12 Act texts available.

Bill (strongest cases)	House: agreed / checked	Act: in the Act / checked
Tax Procedures (Amendment) (No. 2) Bill, 2024	43 (+1 as supported) / 55	47 / 55
Business Laws (Amendment) Bill, 2024	11 / 11	10 / 11
Coffee Bill, 2023 (Senate)	debate not on Mzalendo	21 / 26
Insurance Professionals Bill, 2024	13 / 19	0 / 19 (no bill as published to compare)
Virtual Asset Service Providers Bill, 2025	3 / 41	10 / 41

Strictness was tested and adjusted. The first line-based version confirmed 25 of 41 VASP recommendations. Spot checks found false positives, for example a request to replace the Capital Markets Authority with a single regulator, marked as in the Act because s. 5 differs slightly from the bill. Adding the second check removed these. Allowing longer line ranges (60 Hansard lines, 25 Act lines) then cut House "could not confirm" results caused only by long amendments. Counts move between runs because the model's answers vary. Across the three runs with all checks in place, VASP gave 8, 18 and 10 confirmations. Tax Procedures (41, 45, 47) and Business Laws (8, 9, 10) varied less. Since the checks can only downgrade a claim, variation adds "could not confirm" rather than false confirmations. The figures above are the stored results the pages show.

4. Hosted state. Checks on 6–7 November found the hosted API and schema on the previous release. After a `pg_dump` backup, a cleanup removed demo data and resealed the chain; 10 of 10 receipts verify. **This release is not yet deployed.** Deployment needs:

- `alembic upgrade head` (four new migrations);
- the sync, followed by `analyse_reports --passed`, `trace_house --passed`, and `trace_acts --find` plus Act uploads, run with `GEMINI_API_KEY` set;

- a frontend redeploy.

Next steps: moderator sign-off for AI readings, more Act texts, a second labeller for matching, and a second usability round with individual sessions and timings.

4 Code Quality Practices

Concerns are separated: typed pages and shared components for presentation, `src/api/` for HTTP, routes for coordination, models for persistence, and services for domain logic. Rules that both sides need are defined once per side with the same vocabulary: the submission stages, the participation reason, and the trace statuses. Tests sit at the trust boundaries: chain integrity, access scope, rate limits, untrusted parser input, and AI output. AI output is never trusted directly. Its evidence is taken from the source text and re-checked by deterministic rules before it is shown, and every check has a test. Sync code is defensive: official columns are read before free text, ambiguous rows are skipped rather than guessed, and bills are matched to Acts and Hansard by official number first, then exact title.

Quality risks remain:

- Large modules: `routes.py` has grown to about 2,250 lines and should be split by domain.
- Bundle and start-up size: the bundle is large, and the app loads about 2.3 MB of data at start-up.
- Scaling: rate-limit counters live in memory, so they need Redis before the API runs as several instances.
- Dependencies: the npm advisories are still open.
- AI readings: report, House and Act readings are checked automatically but not signed off by a person; results vary between runs; and scanned Acts and bills (Privatization) depend on OCR quality.
- Sources: Act texts need manual downloads, and House decisions depend on Mzalendo's Hansard copy.
- Matching accuracy: measured on 60 constructed, single-labeller test submissions; it needs a second labeller, real submissions and better Kiswahili handling.

Evidence basis: repository source and tests at 13 November 2026; local Docker Compose runs (backend tests and coverage, Playwright checks, data syncs against parliament.go.ke, county assembly sites and Kenya Law's public listing, Hansard reads from mzalendo.com, Gemini runs of the report, House and Act checks); Act PDFs downloaded from Kenya Law in a browser; hosted HTTP and database checks, and the hosted cleanup, on 6–7 November 2026. Counts and timings are point-in-time observations.